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Effects of high magnetic field on the evolutions of constituent phases in 7085 aluminum alloy during homogenization

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ABSTRACT

The evolutions of coarse constituent phases in homogenized 7085 aluminum alloy at different conditions with or without the application of 12 T high magnetic field were examined by using differential scanning calorimetry, scanning electronic microscope, energy dispersive spectroscopy and X-ray diffraction. It is found that the main constituent phases including quaternary phase T(AlZnMgCu), Al_7Cu_2Fe, and AlTiCuFeSi are present in as-cast 7085 alloy. During homogenization, $\alpha+T$ eutectics become discontinuous and spheroidized, and Al_2CuMg phase nucleates and grows along $\alpha+T$ eutectics. High magnetic field significantly accelerates the melting of quaternary phase T and Al_2CuMg phase. When the alloy homogenized at 460 °C/10 h + 480 °C/8 h with 12 T magnetic field, the least amount of constituent phases is obtained.

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1. Introduction

7085 Aluminum alloy was introduced a few years ago by Alcoa as the next generation of high strength thick plate alloy [1], having very high strength and toughness along with very low sensitivity to gage, and is attributed to its high ratios of Zn/Mg, Cu/Mg and alloying element content, in turn, leading to the difficulty of processing the alloy. The constituent phases, i.e., S (Al_2MgCu), η ($MgZn_2$) or M (Mg (Zn_2AlCu)) and T ($Al_{32}(Mg,Zn)_{49}$), are commonly observed in 7000 series alloy [2]. Many soluble constituents still remain in the alloy after subsequent heat treatments and processing, owing to the proximity of composition to the limit of solid solubility in these alloys [3], which can deteriorate the age hardenability, aid crack initiation and propagation and cause variable properties [4]. K.H. Chen et al. [5] revealed that η phase dissolved completely and T and S phases still remained in 7055 alloy after pretreatment at 450 °C for 35 h. A phase transformation of primary particle from Mg $(Zn,Cu,Al)_2$ phase to Al_2CuMg phase was found in Al–Zn–Mg–Cu alloy during homogenization at 460 °C [6].

The magnetic field, as an effective external field, was verified to have effects on ferromagnetic and non-ferromagnetic materials, and provides us a new approach to material preparation and property control. Many works investigated the influences of magnetic field on the phase transformation of ferrous alloys: (1) phase stability [7–10], (2) phase morphology [11,12], (3) grain refinement [13]; less attentions were paid on aluminum alloy [14–17]. Molodov et al. [14] found that the recrystallization incubation time significantly decreased in cold-rolled 3103 alloy by the application of a 17 T high magnetic field. X. T. Liu et al. [15] discovered that the alternating magnetic field accelerated the nucleation and growth of intermetallic compounds in Al–Cu diffusion couple. W. Liu et al. [16,17] found that the electric field promoted the dissolution of second phase and the removing of interdendritic segregation during homogenization, and even suppressed the nucleation of δ′ phase during artificial aging. In present work, the microstructural changes of 7085 alloy during high magnetic field homogenization were studied in detail by differential scanning calorimetry (DSC), scanning electronic microscope (SEM), energy dispersive spectroscopy (EDS) and X-ray diffraction (XRD) analysis.

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2. Experimental Procedures

7085 Alloy used in present work is Al–7.5wt.%Zn–1.5wt.%Mg–1.65wt.%Cu–0.12wt.%Zr, and was produced from semi-continuous cast ingots with size of Φ100 mm × 1000 mm. Two types of homogenization were conducted with or without the application of 12 T high magnetic field: (1) 460 °C/12 h, air cooling, (2) 460 °C/10 h + 480 °C/8 h, air cooling. The heating rate is 5 °C/min. The temperature variation was controlled within +1 °C. The homogenizations without magnetic field were carried out on a Lenton AWF12/12 type furnace. The apparatus of high magnetic field is illustrated in Fig. 1, which consists of a superconducting magnet and a vacuum resistance furnace. The superconducting magnet can generate an axial static uniform magnetic field with a maximum magnetic flux density of 12 T. Samples were put into a crucible with size of Φ10 mm × 50 mm at center of magnetic field. A stable magnetic intensity of 12 T was applied during homogenization.

Samples for microstructural observations were prepared by standard metallographic method and chemically etched in a solution of 1 ml HF + 1.5 ml HCl + 2.5 ml HNO_3 + 95 ml H_2O, and then examined on a Jeol SSX550 type scanning electron microscopy equipped with a DX-4EDAX type energy dispersive X-ray analyzer. Differential canning calorimetry (DSC) analyses were carried out in a purified argon atmosphere using MDSCQ100 instrument with a scanning rate of 10 °C/min from room temperature to 550 °C. The heat effects associated with transformation reactions were isolated by subtracting a baseline of a high purity aluminum–aluminum run. Phase identifications were examined by X-ray diffraction (XRD) with $CuK\alpha_1$ radiation on PW3040/60X diffractometer.

3. Results and Discussion

DSC curves of 7085 alloy after homogenized at different conditions with or without the application of high magnetic field are illustrated in Fig. 2. Two exothermic peaks (labeled as *a* and *b*, respectively) happened at 476 °C and 492 °C, respectively, are observed in DSC curve of as-cast 7085 alloy. The exothermic peak *a* is ascribed to the dissolution of eutectic α + AlZnMgCu, and the exothermic peak *b* is caused by the melting of Al_2CuMg (S) [6].

After the alloy homogenized at 460 °C/12 h, peak *a* reduces indicating the decrease in amount of α + AlZnMgCu, while, peak *b* enlarges demonstrating the increase in amount of S phase. It can be seen that the sizes of peak *a* and *b* decrease with the application of stepped homogenization and high magnetic field, and become the smallest when the alloy stepped homogenized at 460 °C/10 h + 480 °C/8 h with 12 T high magnetic field.

The XRD analysis results of 7085 alloy after homogenized at different conditions with or without the application of high magnetic field are shown in Fig. 3. The microstructure of as-cast 7085 alloy contains $MgZn_2$ and Al_7Cu_2Fe. No peak associated with S phase can be observed in as-cast 7085 alloy. The peak of S phase arises in the alloy homogenized at 460 °C for 12 h. The amount of the peak of S phase decreases further when the alloy homogenized at 460 °C/10 h + 480 °C/8 h, and becomes the least when the alloy homogenized at 460 °C/10 h + 480 °C/8 h with 12 T high magnetic field.

The as-cast microstructure of 7085 alloy (Fig. 4) reveals a mixture of dendritic α-Al and lamellar eutectic formed between dendrites. From EDS results, point 1 is the quaternary phase AlZnMgCu and has a composition of 4.69 wt. % Mg, 40.38 wt. % Cu, 48.62 wt. % Zn and 6.31 wt. % Al. The point 2 contains 1.98 wt. % Mg, 15.93 wt. % Al, 35.26 wt. % Cu, 7.24 wt. % Zn, 33.13 wt. % Fe, 6.46 wt. % Si and 1.28 wt. % Ti. Because of the small size of phase 2, the electric beam is easy to detect the matrix during the procedure of EDS analysis, and then the composition of matrix (point 3) is given as: 1.27 wt. % Mg, 87.08 wt. % Al and 11.65 wt. % Zn. Thus, the phase 2 is determined to be AlTiCuFeSi. The point 4 is Al_7Cu_2Fe and has a composition of 19.56 wt. % Al, 34.52 wt. % Cu, 7.18 wt. % Zn and 38.74 wt. % Fe. It should be noted that plenty of eutectic α + AlZnMgCu instead of η are observed in as-cast microstructure of 7085 alloy; it is because that the high cooling speed during semi-continuous casting causes the non-equilibrium solid solution of element Al and Cu in η phase and thus results in the formation of eutectic α + AlZnMgCu. It is reported that S, T and η are solid solutions with extended composition ranges [4], so the quaternary phase AlZnMgCu observed in present work belongs to T phase.

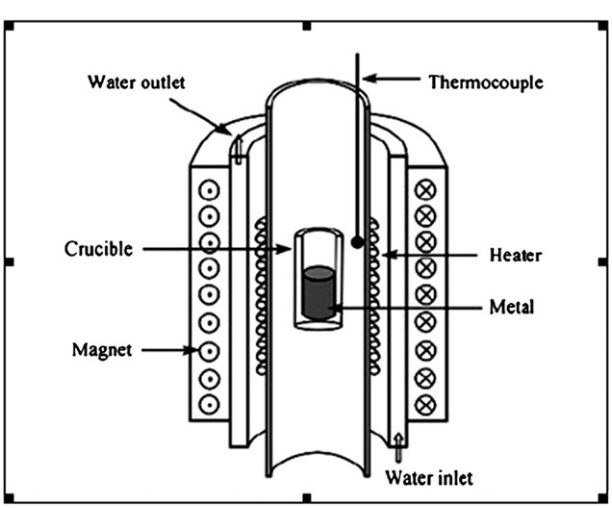

Fig. 1 – Schematic diagram of high magnetic field system.

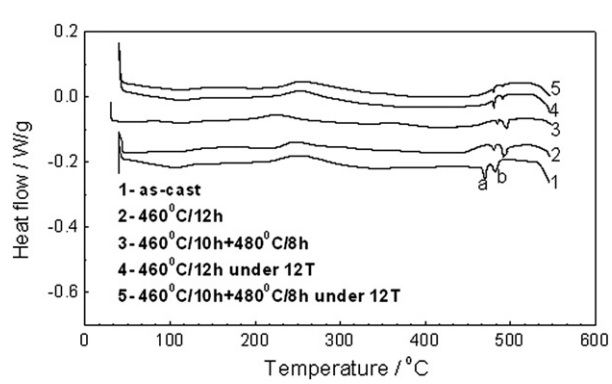

Fig. 2 – DSC curves of 7085 alloy homogenized at different conditions.

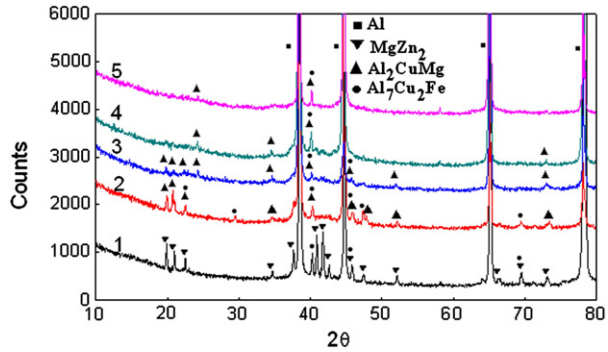

Fig. 3 – XRD results of 7085 alloy homogenized at different conditions: 1-as-cast, 2–460 °C/12 h, 3–460 °C/10 h + 480 °C/8 h, 4–460 °C/12 h under 12 T high magnetic field, and 5–460 °C/10 h + 480 °C/8 h under 12 T high magnetic field.

After the alloy homogenized at 460 °C/12 h (Fig. 5), the coarse eutectics become smaller, discontinuous and spheroidized, and appears as a necklace-like distribution at grain boundaries. According to the EDS results, besides Al_7Cu_2Fe (point 5) and eutectic α + T (point 6), a new phase (point 7) having a dark gray is detected at the margin of eutectic α + T and contains 20.97 wt. % Mg, 47.82 wt. % Al, 24.78 wt. % Cu, and 6.43 wt. % Zn, while the matrix (point 8) contains 1.32 wt. % Mg, 78.5 wt. % Al and 20.18 wt. % Zn, so it can be determined that the phase 7 is Al_2CuMg (S) phase. The black bots in Figs. 6–8 are the eutectic α + T falling off during the procedure of sample preparation. The size and the amount of eutectic α + T and S phase decrease further with the application of step homogenization and high magnetic field (Figs. 6–8).

The observed phases in as-cast microstructures in 7085 alloy are eutectic α + T and Fe-bearing phases (AlTiCuFeSi or Al_7Cu_2Fe). It should be pointed out that S phase could not be observed in as-cast microstructure by using SEM, EDS and XRD measurements, which is different from the results made by Xie et al. [18]. This may result from the differences in alloy composition or casting conditions. The detected exothermic peak caused by the melting of S phase in DSC curve of as-cast 7085 alloy could result from the formation of S phase during the heating procedure of DSC experiment [19]. During

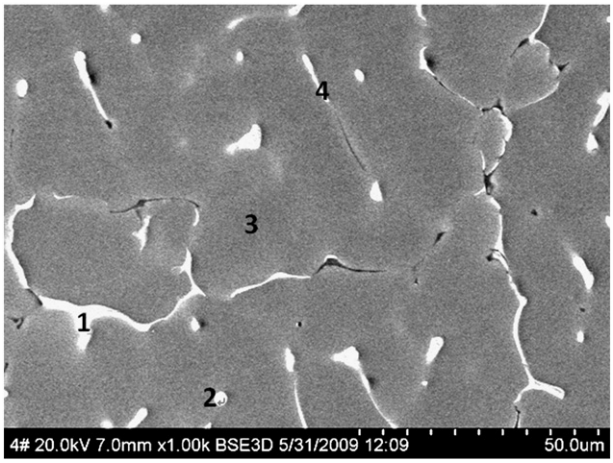

Fig. 4 – SEM microstructures of as-cast 7085 alloy.

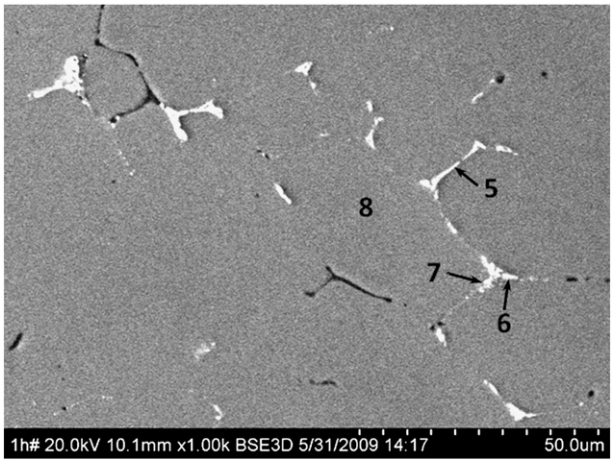

Fig. 5 – SEM microstructures of 7085 alloy homogenized at 460 °C/12 h.

homogenization, the eutectics α + T dissolve into the matrix, and at the same time, Al_2CuMg phase forms at the base of α + T which indicates that the transformation from T to S phase occurs. The alloying elements in the eutectic structures diffuse into the matrix, so the T phase dissolves gradually during homogenization. The diffusion velocity of Cu is lower than Zn and Mg, which results in the higher concentration of Cu in these regions. The driving force for the phase transformation from T phase to S phase must be the supersaturation of Cu in eutectic structures, which makes S phase a stable phase. The size and amount of AlZnMgCu and S phase decrease further after the alloy homogenized at 460 °C/10 h + 480 °C/8 h, indicating the dissolution of the newly formed S phase. This is due to the diffusion of alloying elements from the interdendritic regions to the matrix, and thus decreases the supersaturation extent of alloying elements in eutectic regions. Thus, the evolution sequence of constituent phases in 7085 alloy during homogenization can be described as follows: the dissolution of eutectics α + T, the transformation from T to S, and the dissolution of S.

With the application of 12 T high magnetic field, the amount of T and S significantly decreases comparing to the

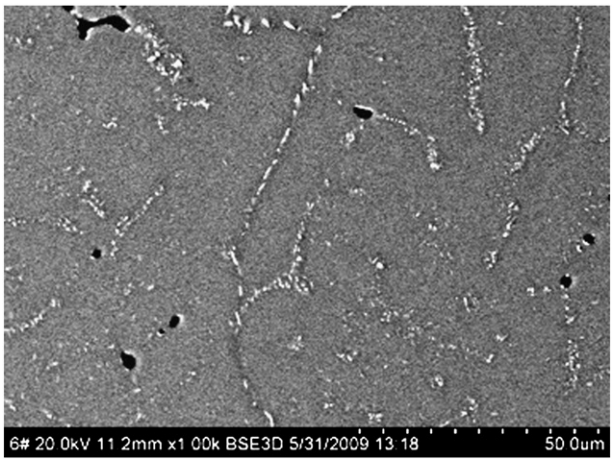

Fig. 6 – SEM microstructures of 7085 alloy homogenized at 460 °C/10 h + 480 °C/8 h.

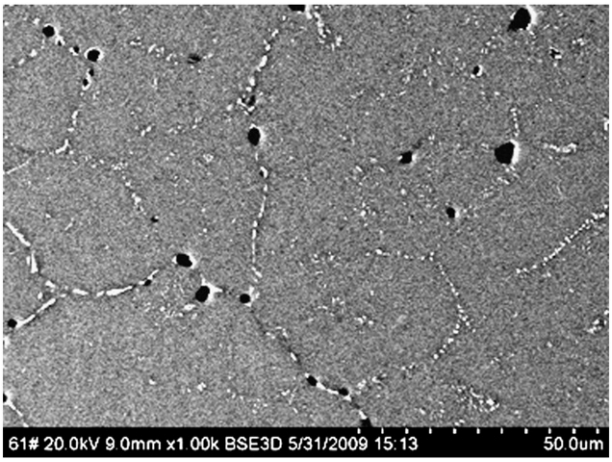

Fig. 7 – SEM microstructures of 7085 alloy homogenized at 460 °C/12 h under 12 T high magnetic field.

conventional homogenization with the same conditions, which demonstrates that the high magnetic field can effectively accelerate the dissolution rate of T and S phase. In conformance with the transportation theory of plasma, mass transport in metals and alloys by chemical diffusion is an ambipolar process [20]. The diffusion processes of ions and electrons move at different speeds in a metal. A weak electric field couples the processes together to preserve charge neutrality. The electric field in turn affects the transport resulting in deviation from purely diffusive behavior. During the homogenization with the application of high magnetic field, the motion of free electrons includes the gyration under the action of Lorentz force and the drift by the guiding center in the magnetic field, as a result of which the diffusion of the transported electrons is accelerated and ambipolar diffusion process of the atoms in metal is promoted. The magnetic field accelerates the diffusion process and so the dissolution of constituent phases.

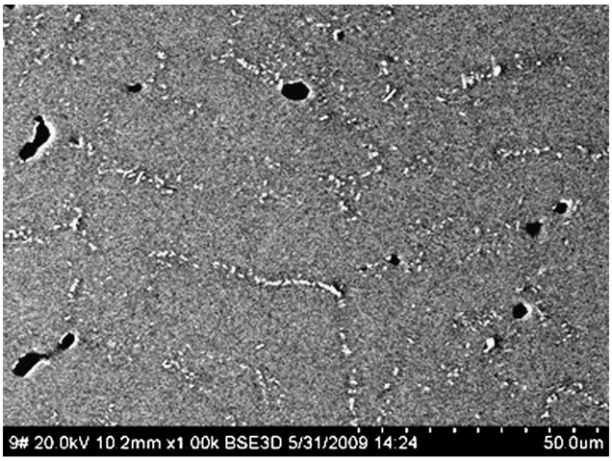

Fig. 8 – SEM microstructures of 7085 alloy homogenized at 460 °C/10 h + 480 °C/8 h under 12 T high magnetic field.

4. Conclusions

The as-cast microstructures of 7085 alloy consist of eutectic α + T, Al_7Cu_2Fe and AlTiCuFeSi. The overheating temperature of 7085 alloy is 476 °C.

During homogenization, the transformation from T to S phase occurs. The amount of T and S phase decreases among the four-homogenization conditions in the following sequence: 460 °C/12 h, 460 °C/10 h + 480 °C/8 h, 460 °C/12 h with high magnetic field, and 460 °C/10 h + 480 °C/8 h with high magnetic field.

The high magnetic field obviously accelerates the dissolution rate of T and S phase during homogenization.

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